AI-Driven Environmental Impact Prediction System for Sustainable Textile Waste Reuse

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Project Proposal Report

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BSc (Hons) in Information Technology Specializing in Information Technology

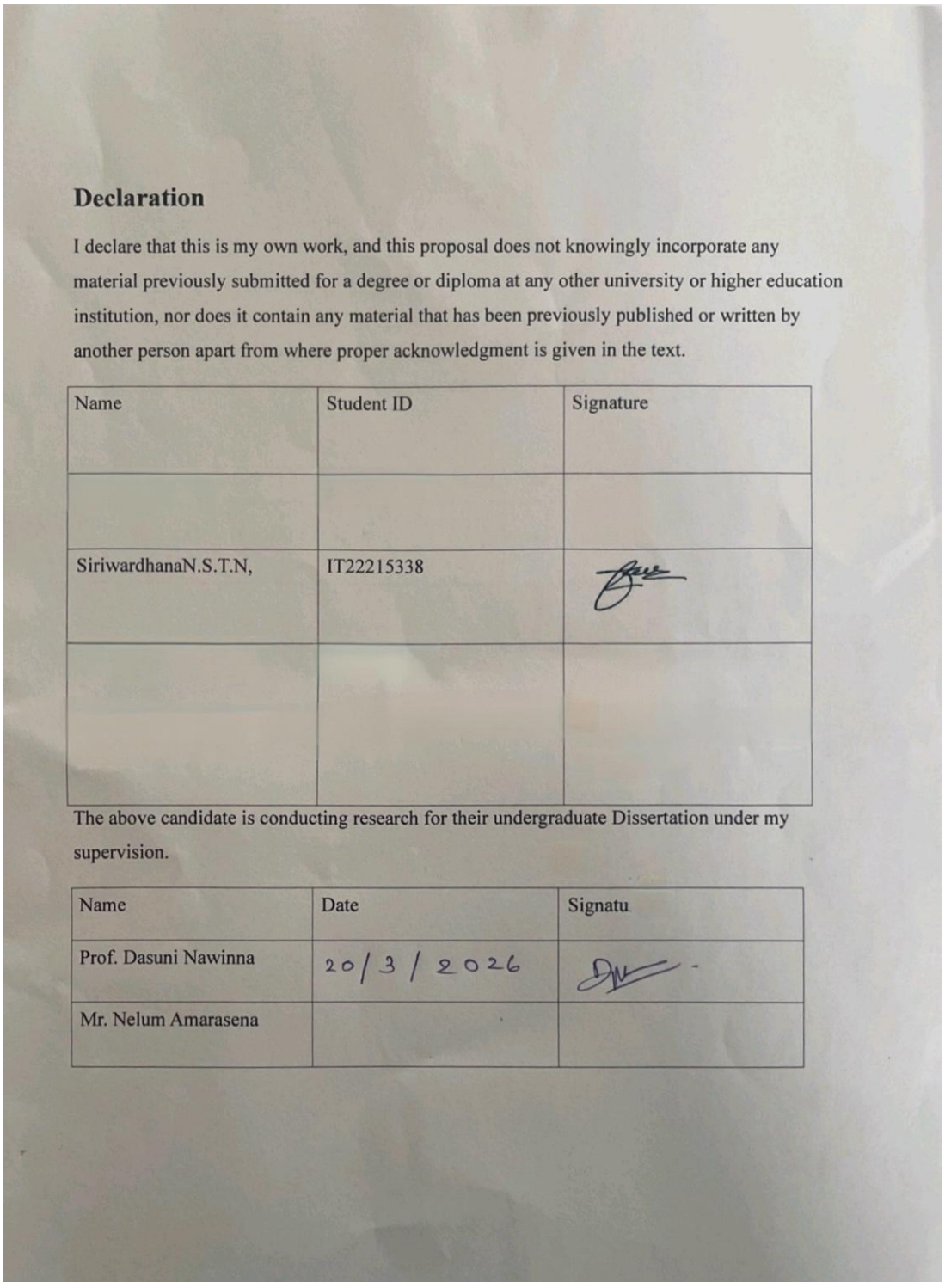
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Declaration



Declaration

I declare that this is my own work, and this proposal does not knowingly incorporate any material previously submitted for a degree or diploma at any other university or higher education institution, nor does it contain any material that has been previously published or written by another person apart from where proper acknowledgment is given in the text.

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The above candidate is conducting research for their undergraduate Dissertation under my supervision.

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Abstract

Sri Lanka's textile industry produces a lot of fabric waste, but the country's present redistribution programs undervalue and fail to quantify the environmental benefits of recycling this waste. This research closes this gap by developing an AI-driven sustainability intelligence module that predicts the environmental impact of repurposed textile waste at the batch level. The proposed system calculates the quantity of water and carbon dioxide (CO_2) that may be saved by reusing garbage using regression modeling approaches, including Random Forest and Linear Regression algorithms. By combining publicly available Lifecycle Assessment (LCA) data with fabric characteristics including material type, weight, and quality grade, the algorithm generates a quantifiable "Green Score" for each trash batch. This score encourages sustainable purchase decisions by giving market players clear, intelligible metrics. The module uses a Fast API microservice to deliver predictions and is made for easy integration into a larger digital marketplace platform. A functional API service, precise environmental impact forecasts (target $R^2 > 0.85$), and a contribution to the advancement of circular economy concepts in Sri Lanka's textile industry erectly supporting SDGs 12 (Responsible Consumption and Production) and 13 (Climate Action), are among the anticipated results.

Keywords: Lifecycle Assessment (LCA), Water Footprint, CO2 Savings, Machine Learning, Circular Economy, Sustainability Intelligence, Regression Modeling, Environmental Impact Prediction, Textile Waste, and Green Score.

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List of abbreviations

Abbreviation	Full Form
AI	Artificial Intelligence
API	Application Programming Interface
CO_2	Carbon Dioxide
CSV	Comma-Separated Values
DSR	Design Science Research
EPLCA	European Platform on Life Cycle Assessment
ESG	Environmental, Social, and Governance
LCA	Life Cycle Assessment
MAE	Mean Absolute Error
ML	Machine Learning
MSI	Materials Sustainability Index
RMSE	Root Mean Square Error
R^2	Coefficient of Determination
SDG	Sustainable Development Goals
SME	Small and Medium Enterprise
TIM	Technology Integration and Management
UNEP	United Nations Environment Programme
JSON	JavaScript Object Notation
REST	Representational State Transfer

1. Introduction

1.1 Background and Context

A substantial amount of textile waste is produced by the textile industry, which is one of the biggest manufacturing industries in the world. Although the clothing industry in Sri Lanka plays a significant role in the country's economy, the creation of clothing generates a significant amount of waste fabric. In addition to depleting important natural resources needed for the manufacturing of virgin textiles, disposing of textile waste by landfilling or incineration adds to greenhouse gas emissions, soil contamination, and water pollution [1].

Textile waste has grown to be a significant environmental issue on a global scale. The Ellen MacArthur Foundation estimates that one garbage truck's worth of textiles is burned or landfilled globally every second [2]. This linear "take-make-dispose" production process uses resources inefficiently and is not environmentally sustainable.

Reusing and upcycling textile waste has a major positive impact on the environment. Reusing fabric waste lessens the need to produce virgin materials, which need a lot of water and energy. Additionally, it reduces carbon emissions from manufacturing processes and preserves the integrated energy previously used in textile manufacture [3]. Nevertheless, most designers, producers, and consumers find it difficult to quantify the environmental benefits of reusing textile waste.

Current textile waste trading systems don't offer quantifiable sustainability insights; instead, they mostly concentrate on facilitating transactions between buyers and sellers. Because of this, manufacturers are unable to effectively convey the sustainability benefits of their waste materials, and consumers are unable to assess the environmental value of textile waste products.

This study suggests an AI-driven sustainability intelligence module that can forecast the environmental impact of reusing textile waste at the batch level in order to solve this problem. The technology calculates the amount of water saved and CO2 emissions decreased when textile waste is reused by integrating Lifecycle Assessment (LCA) data with machine learning algorithms. A simplified sustainability indicator called a "Green Score" summarizes these parameters, empowering users to make knowledgeable and ecologically conscious choices.

The proposed system supports the principles of the circular economy**,** encouraging the reuse of textile waste and reducing environmental damage. Additionally, the research aligns with the United Nations Sustainable Development Goals (SDGs)**,** particularly SDG 12 (Responsible Consumption and Production) and SDG 13 (Climate Action)**.** From a technological perspective, the research aligns with the Technology and Information Management (TIM) research cluster**,** integrating artificial intelligence, sustainability analytics, and digital marketplace technologies.

1.2 Literature Review

A common approach for evaluating a product's environmental effects across its whole lifecycle, from the extraction of raw materials to disposal, is Lifecycle Assessment (LCA). LCA allows researchers to measure environmental impacts such as carbon emissions, water use, and energy usage, in accordance with ISO 14040/14044 standards [4].

Several studies have highlighted the environmental impact of textile production. For example, producing one kilogram of conventional cotton requires approximately 10,000–20,000 liters of water and generates considerable greenhouse gas emissions due to fertilizer use and mechanized farming [5]. Similarly, polyester production, although less water-intensive, relies heavily on fossil fuels and energy-intensive manufacturing processes, leading to significant carbon emissions [5].

Characterization factors, which calculate the environmental impact per unit of material, are frequently used by academics to quantify these effects. Well-known databases like the Eco invent database, the European Platform on LCA (EPLCA), and the Higg Materials Sustainability Index (MSI) created by the Sustainable Apparel Coalition are frequently used to get these parameters [6].

In LCA studies, the Avoided Burden Approach is often used to assess the environmental advantages of recycling or material reuse. Reusing a material avoids the need to generate an equivalent amount of virgin material, according to this method. Reusing one kilogram of cotton waste, for example, eliminates the environmental effects of making one kilogram of new cotton fabric [8]. Therefore, the environmental savings can be approximated using the impact values associated with the virgin material it replaces.

The incorporation of machine learning methods for environmental effect prediction has also been studied recently. Strong performance in predicting sustainability indicators in manufacturing and environmental monitoring systems has been shown by machine learning models such Random Forest, Gradient Boosting, and Linear Regression [9].

Despite these developments, integrated environmental effect prediction systems are still absent from current digital platforms for the trading of textile waste. Most platforms just concentrate on waste exchange and don't offer measurable sustainability insights to help consumers make eco-friendly purchases.

1.3 Research Gap

There are still several gaps in the extensive study on environmental effect assessment in the textile sector.

First, rather than estimating environmental savings at the individual textile waste batch level, the majority of life cycle assessment (LCA) studies concentrate on large-scale production analysis. For textile waste transactions, there is yet no automated mechanism that translates LCA data into real-time sustainability measures.

Second, small-scale designers, entrepreneurs, and manufacturers frequently lack the technical know-how and specialized information needed to undertake life cycle assessments. This makes it more difficult to take sustainability into account when making decisions about production and purchases.

Third, existing digital platforms that facilitate textile waste exchange do not integrate environmental impact analytics. As a result, users cannot evaluate the sustainability benefits of textile waste reuse in a data-driven manner

Lastly, environmental effect data is not included into circular economy workflows by existing technologies. The environmental impact of textile waste over the course of its digital lifecycle is not automatically calculated, displayed, or recorded.

To automatically forecast and convey the environmental value of textile waste reuse, a data-driven solution that combines LCA data, machine learning, and digital marketplace technologies is required.

2. Objectives

2.1 Main Objective

To create and implement an AI-driven prediction model that precisely calculates the environmental impact (water conservation and CO2 emission reduction) of reusing textile waste, and to integrate it as a microservice with a digital marketplace platform for textile trash.

2.2 Specific Objectives

1. To create an organized collection of Lifecycle Assessment (LCA) environmental impact indicators for common textile materials, including blended textiles, cotton, polyester, nylon, and viscose.

2. To provide feature engineering methods that convert textile waste attributes (weight, quality grade, and type of fabric) into input features for machine learning.

3. To develop and train machine learning regression models such as Linear Regression, Random Forest, and Gradient Boosting to predict CO_2 emission reductions and water savings from textile waste reuse.

4. Regression evaluation metrics, such as R^2, MAE, and RMSE, are used to assess the constructed models' performance and guarantee a satisfactory level of prediction accuracy.

5. to seamlessly integrate the top-performing prediction model with a digital marketplace platform by implementing it as a RESTful API microservice using Fast API.

6. To create a Green Score sustainability indicator that incorporates environmental effect forecasts into an easy-to-understand sustainability rating for users

3. Methodology

3.1 Research Approach

System integration, model engineering, and data collection will all be part of the structured, iterative approach used to construct this module. Data collection and sourcing, feature engineering and preprocessing, baseline model development, machine learning model development, model evaluation and selection, and API development and deployment comprise the six primary stages of the technique. Below is a detailed description of each phase.

3.2Data Collection and Ethical Considerations

The research will rely on secondary data from reliable, publicly accessible sources because primary LCA data collection, conducting original, full-scale LCAs, is outside the purview of this undergraduate project.

- **LCA Databases:** Characterization factors will be taken from reputable LCA databases, such as the European Commission's European Platform on LCA (EPLCA) and the Eco Invent database (accessible under academic licenses or papers citing it).
- **Academic Literature:** To identify pertinent impact factors, peer-reviewed life cycle assessment (LCA) research on textile fibers (such as cotton, polyester, and wool) will be examined. Average values from meta-analyses will be especially useful.
- **Industry Frameworks:** Data from the Sustainable Apparel Coalition's Higg Materials Sustainability Index (MSI) will be used, as it provides readily accessible, industry-vetted impact scores for a wide range of materials.
- **Dataset Creation:** A structured dataset will be created in CSV format. Each row will represent a material type, with columns for:
    - Material name (such as "Cotton, conventional")
    - Type of material (such as "Natural cellulosic")
    - CO2 factor kg per kg (kg CO2 equivalent saved per kg of material)
    - water factor l per kg (liters of water saved per kilogram of material)
    - source (data citation)

Table 1: Sample LCA Factors for Common Textile Materials (Illustrative).

Material Name	CO_2 Factor (kg CO_2 eq./kg)	Water Factor (L/kg)	Source (Example)
Cotton (Conventional)	5.0	10,000	[6]
Polyester (Virgin)	7.5	100	[7]
Nylon 6,6	9.0	150	EcoInvent v3.8
Viscose (Generic)	4.5	600	Higg MSI

3.3 Technical Methods and Algorithms

The model's input features are made to be easy to collect from other system modules (like the fabric categorization module) or from the user (seller).

Primary Features:

- **fabric type (Categorical):** The type of fabric (e.g., Cotton, Polyester, Blend). This will be the primary key to link to the LCA factor database.
- **weight -key (Numerical):** The total weight of the waste batch in kilograms.
- **quality grade (Categorical/Ordinal):** A straightforward grade that could serve as a discount factor, such as High, Medium, or Low.
- For instance, the savings could be modified (e.g., savings = virgin factor * weight * quality multiplier, where High=1.0, Medium=0.8, Low=0.5) if low-quality cloth merely displaces a portion of virgin material.

Derived Features / Target Variables:

- **total_co2_saved (Numerical):** The CO2 model's target output. computed using the formula co2_factor * weight * quality multiplier.
- **Total water saved (Numerical):** The water model's desired result. Water factor * weight * quality multiplier is the calculation.
- **green score (Numerical/Categorical):** A composite score, possibly an A-F grade or a normalized average of the water and CO2 savings (e.g., on a scale of 1-100).

3.4 Technical Methods and Algorithms (Model Development)

3.5 Validation Strategy and Performance Metrics

- **Data Splitting:** Training (80%) and testing (20%) sets will be created from the synthetic dataset.
- **Training:** To reduce prediction error, the regression models will be trained on the training set.
- **Validation:** Standard regression metrics will be used to assess the model's performance on the unseen test set.
- **R-squared (R^2):** The primary metric, measuring how well the model explains the variance in the data. Target > 0.85.
- **Mean Absolute Error (MAE):** The average absolute difference between predicted and actual values. Lower is better.
- **Root Mean Square Error (RMSE):** Like MAE but gives more weight to large errors.
- **Model Selection:** The best-performing model will be deployed based on its interpretability and validation criteria. The direct computation approach will be the last option if it turns out to be sufficiently precise and is favored due to its transparency.

3.6 API Development and Deployment

The final model will be deployed as a microservice using Fast API, a modern, high-performance Python web framework.

- Endpoint: A single POST endpoint, e.g., /predict impact.
    - Request

      Body:



```
{
  "fabric_composition": [{"type": "Cotton", "percentage": 60}, {"type": "Polyester", "percentage": 40}],
  "weight_kg": 5.5,
  "quality_grade": "High"
}
```

Figure 1: Request JSON Format.

    - Response

      Body:



```
{
  "co2_saved_kg": 27.5,
  "water_saved_l": 55000,
  "green_score": "A",
  "message": "Success"
}
```

Figure 2: Response JSON Format.

- Error Handling: The API will include robust error handling for invalid inputs or missing fabric types.
- Documentation: Fast Api's automatic interactive API documentation (Swagger UI) will be generated, making it easy for other team members to integrate the module.

4. High-level system architecture

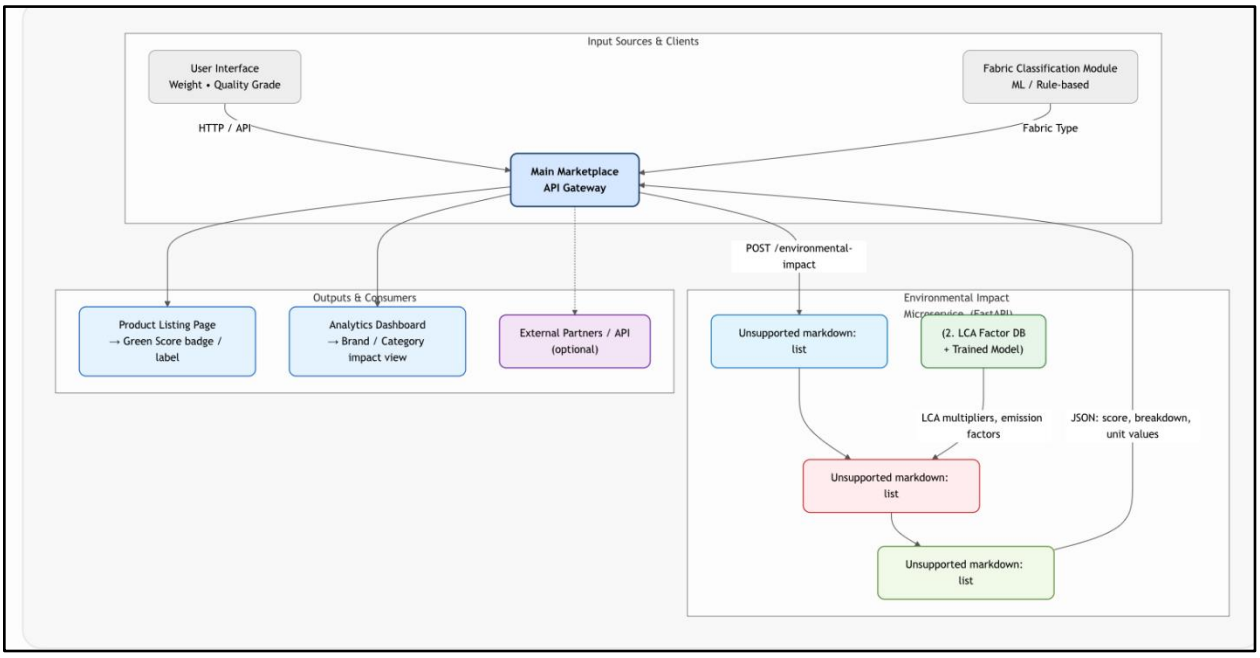


Figure 3: System Overview Diagram – Fabric Classification Module

5.Requirements

5.1 Functional Requirements

Table 2: Functional Requirements Table.

ID	Requirement Description
FR-01	The system shall accept input parameters: fabric type (string), weight_kg (float), and quality grade (string).
FR-02	The system shall maintain a database or lookup table of LCA factors (CO_2/kg, Water/L/kg) for at least 15 common textile materials.
FR-03	The system shall calculate the total estimated CO_2 emissions saved (in kg) based on the input and LCA factors.
FR-04	The system shall calculate the total estimated water volume saved (in liters) based on the input and LCA factors.
FR-05	The system shall generate a composite "Green Score" (e.g., A–F grade or 1–100 score) based on the combined CO_2 and water savings.
FR-06	The system shall return the calculated metrics in a structured JSON format via a RESTful API endpoint.
FR-07	The system shall handle invalid inputs (e.g., unsupported fabric type, negative weight) gracefully and return an informative error message.
FR-08	The system shall log all prediction requests for performance monitoring and future model retraining.

5.2. Non-Functional Requirements

Table 3: Non-Functional Requirements Table.

ID	Requirement Description
NFR-01	Accuracy: The model's predictions for CO_2 and water savings shall have an R^2 value of >0.85 when compared against the baseline LCA-derived values.
NFR-02	Performance: The API shall respond to prediction requests in under 500 milliseconds under normal load.
NFR-03	Scalability: The microservice shall be designed to handle multiple concurrent requests (tested with up to 50 simultaneous users).
NFR-04	Reliability: The service shall have 99.5% uptime during development and testing phases.
NFR-05	Usability: The output "Green Score" must be easily understandable by a non-expert user.
NFR-06	Maintainability: The codebase shall be well-documented, and the LCA factor database shall be easily updatable as new data becomes available.
NFR-07	Transparency: The system should be able to explain the source of its LCA data upon request (e.g., by providing the data source in the API response or documentation).

5.3 System Requirements

- **Programming Language:** Python 3.9+
- **Web Framework:** Fast API
- **Machine Learning Libraries:** scikit-learn, pandas, NumPy (for model training and inference)
- **Deployment:** Docker container for a consistent environment; deployed on a cloud VM (e.g., AWS EC2, Google Cloud Run) or as part of the main platform's backend.
- **API Communication:** REST over HTTP/HTTPS.
- **Data Storage:** The LCA factor database can be stored as a simple CSV file or in a lightweight database like SQLite for the prototype; for production, it could be moved to the main MongoDB instance.
- **Version Control:** Git (GitHub/GitLab).

5.4 User Requirements

Users are not expected to interact with this module directly. It is a backend service. However, users interacting with the main marketplace platform implicitly require.

- Basic understanding of sustainability metrics (e.g., knowing that a lower carbon footprint is better).
- Ability to interpret the Green Score displayed on product listings.

5.5 Use Cases

Table 4: Use Case 1 – Use Cases.

Element	Description
Use Case ID	UC-01
Name	Generate Environmental Impact Prediction
Summary	This use case describes how the system automatically calculates and returns the environmental impact metrics for a batch of textile waste.
Priority	High
Preconditions	• The main marketplace platform has received a request to create a new product listing. • The fabric type, weight, and quality grade for the waste batch have been provided (by user or classification module).
Postconditions	• The environmental impact metrics (CO_2 saved, water saved, Green Score) are calculated and returned to the calling platform.
Primary Actor(s)	Main Marketplace Platform (System)
Secondary Actor(s)	None
Trigger	The main platform's API sends a POST request to the **/predict impact** endpoint upon receiving a new waste batch listing.
Main Scenario	1. The main platform sends a request with fabric type, weight (kg), and quality grade. 2. The microservice validates the input data. 3. The service retrieves the corresponding LCA factors from its database. 4. The service calculates **total CO_2 saved** and **total water saved**. 5. The service calculates the **composite Green Score**. 6. The service packages the results into a JSON response. 7. The service returns the JSON response to the main platform with **HTTP status 200**.

Extensions	• **E1 (Invalid Input):** If the fabric type is not found in the database, the service returns an error message like {"error": "Unsupported fabric type"}. • **E2 (Missing Input):** If required fields are missing, the service returns a **422 Unprocessable Entity** error.
Open Issues	• How to accurately handle blended fabrics (e.g., 60% cotton, 40% polyester). This may require a **weighted average calculation** based on blend percentage.

Table 5: Use Case 2 –View Batch-Level Sustainability Metrics

Element	Description
Use Case ID	UC-02
Name	View Batch-Level Sustainability Metrics
Summary	A buyer browsing the marketplace can view the calculated environmental impact metrics for a specific waste batch on its product detail page.
Priority	Medium
Preconditions	• The environmental impact module has successfully generated metrics for the batch (UC-01). • The metrics are stored in the main database associated with the product listing. • The user is logged into the platform.
Postconditions	• The user views the environmental impact data for the batch.
Primary Actor(s)	Buyer (Job Seeker / Designer / Upcycler)
Secondary Actor(s)	Marketplace Platform
Trigger	The user navigates to the product detail page for a specific waste batch listing.

Main Scenario	1. The user clicks on a product listing from a search result or dashboard. 2. The marketplace platform retrieves the product data, including the **pre-calculated CO_2 saved (kg), water saved (L), and Green Score**. 3. The platform renders the product detail page. 4. The user sees the metrics displayed prominently, possibly with visual aids (e.g., "This batch saves CO_2 equivalent to driving X km in a car").
Extensions	**E1:** The user can click on a **"Learn More" or "Methodology"** link to understand how the metrics were calculated and the source of the LCA data.
Open Issues	• How to present the metrics most engagingly and understandably for users with varying levels of sustainability knowledge.

5.6 Wireframes

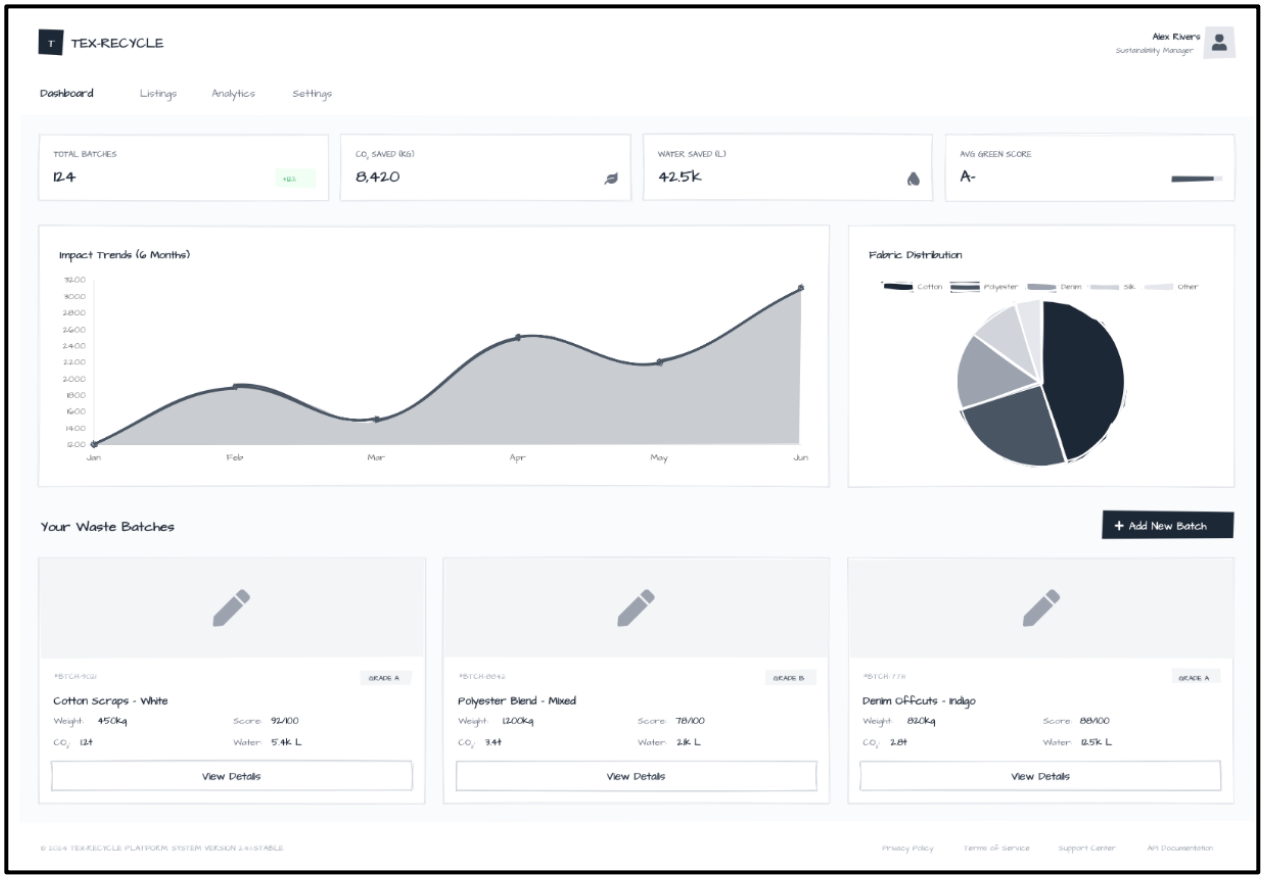


Figure 4: Wireframe – Impact Dashboard Wireframes.

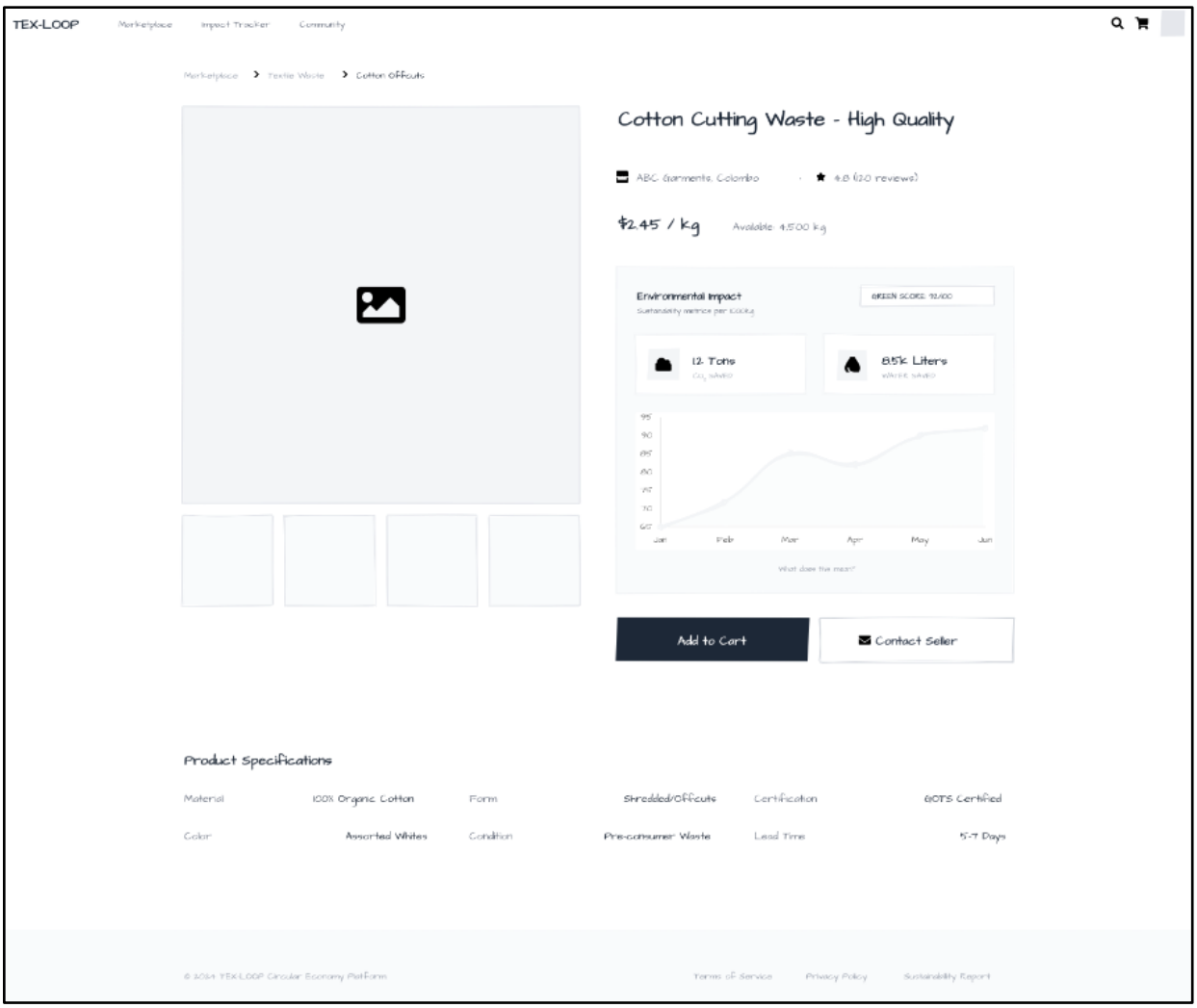


Figure 5: Wireframe – Cotton Cutting Waste Wireframes.

6. Commercialization Plan

6.1 Target Market

- Primary: A key component of the broader "Text read" marketplace is the module. The platform's buyers and sellers are its end users.
- Secondary: The module may be licensed as a stand-alone "Sustainability Analytics API" for corporate sustainability reporting tools, waste management firms, or other HR platforms.
- Market Space: The worldwide market for ESG (Environmental, Social, and Governance) software and sustainability reporting is expanding quickly. This module focuses on the South Asian textile and apparel value chain, which is a particular specialization within that industry.
-

6.2 Value Proposition:

- For Manufacturers (Sellers): Enables them to market waste with verified environmental credentials, potentially commanding a "green premium" and improving their own ESG reporting.
- For Designers/Upcycles (Buyers): Provides data to make informed, sustainable purchasing decisions and to credibly communicate the environmental benefits of their own products to their customers.
- For the Platform: Creates a key differentiator from generic marketplaces and positions the platform as a leader in circular economy technology.

How to Promote:

- Content Marketing: Publish white papers and case studies on the methodology and its application, targeting sustainability professionals in the apparel industry.
- Partnerships: Collaborate with industry bodies like the Joint Apparel Association Forum (JAAF) in Sri Lanka to showcase the tool as a resource for member companies.
- Freemium Model: Offer basic impact calculations for free on the platform, with more detailed analytics and reporting features available as a premium subscription.

7. Budget and budget justification

Table 6: Budget Table.

Component	Amount (USD)	Amount (LKR)	Justification
Data and Software			
Access to LCA Databases (if required)	0.00	0.00	Using publicly available/open-access data from literature and free industry resources (e.g., Higg MSI public data).
Cloud Computing Credits (e.g., Google Cloud, AWS)	30.00	9,000.00	For model training (GPU not strictly required but helpful), API deployment, and load testing. Free tiers may cover initial needs.
Infrastructure and Tools			
Development Tools (IDE, Git)	0.00	0.00	All tools are open-source or have free educational licenses.
Internet Charges (Development & Deployment)	15.00	4,500.00	For research, cloud access, and collaborative development over 5 months.
Knowledge and Consultation			
Technical Consultation (AI/ML Specialist)	20.00	6,000.00	For expert guidance on model selection, hyperparameter tuning, and validation methodology.
Industry Consultation (Sustainability Expert)	15.00	4,500.00	For validation of LCA data sources and interpretation of environmental metrics.
Contingency			
Miscellaneous / Contingency Fund	10.00	3,000.00	For unforeseen costs (e.g., higher than expected cloud usage, printing of reports).
TOTAL	**90.00**	**27,000.00**	

8. Work breakdown structure.

Table 7: Work Breakdown Table

WBS ID	Task / Component	Description
1.0	Environmental Impact Module (Siriwardhana N.S.T.N.)	The main module is responsible for calculating and predicting environmental impact metrics.
1.1	**Research & Planning**	**Initial research and requirement analysis phase.**
1.1.1	Conduct LCA Literature Review	Review existing research on Life Cycle Assessment (LCA) and textile environmental impacts.
1.1.2	Identify Key Textile Materials & Impact Factors	Determine materials (cotton, polyester, etc.) and their environmental impact factors.
1.1.3	Document System Requirements	Define system requirements and functional specifications.
1.2	**Data Engineering**	**Collecting and preparing data for the model.**
1.2.1	Source LCA Data from Databases/Literature	Gather environmental impact data from LCA databases and research papers.
1.2.2	Create Structured LCA Factor Dataset (CSV)	Organize collected data into a structured dataset format.
1.2.3	Develop Synthetic Training Dataset	Generate synthetic data to support model training.
1.3	**Model Development**	**Implementation of prediction models.**
1.3.1	Implement Baseline Lookup Model	Create a simple rule-based model using LCA factors.
1.3.2	Implement Linear Regression Model	Develop a linear regression model for prediction.
1.3.3	Implement Random Forest Regressor	Implement a Random Forest machine learning model.

1.3.4	Implement Gradient Boosting (XGBoost)	Implement an advanced boosting algorithm for improved predictions.
1.4	**Model Validation**	**Evaluate model performance and accuracy.**
1.4.1	Split Data into Train/Test Sets	Divide the dataset into training and testing subsets.
1.4.2	Train All Models	Train the developed models using training data.
1.4.3	Evaluate Performance (R^2, MAE, RMSE)	Measure prediction accuracy using evaluation metrics.
1.4.4	Select and Tune the Best Model	Choose the best-performing model and optimize its parameters.
1.5	**API Development**	**Build the backend service for model predictions.**
1.5.1	Set Up FastAPI Project Structure	Initialize project structure using FastAPI framework.
1.5.2	Implement /predict impact Endpoint	Create an API endpoint to calculate environmental impact.
1.5.3	Implement Input Validation & Error Handling	Ensure proper input validation and error responses.
1.5.4	Write Unit & Integration Tests	Develop tests to verify API functionality.
1.6	**Deployment & Integration**	**Deploy and integrate the system.**
1.6.1	Containerize with Docker	Package the application into Docker containers.
1.6.2	Deploy Microservice to Cloud/VM	Deploy the API service to a cloud or virtual machine environment.
1.6.3	Test API Integration with Main Platform	Ensure proper communication between the API and the main platform.
1.6.4	Document API Usage	Provide documentation for developers and users.
1.7	**Reporting & Documentation**	**Final documentation and presentation.**
1.7.1	Write Methodology Section (Final Paper)	Document the system design and research methodology.

1.7.2	Write Results Section (Final Paper)	Present experimental results and model performance.
1.7.3	Prepare Project Demonstration	Create a presentation and demo for the final evaluation.

9. Gantt chart

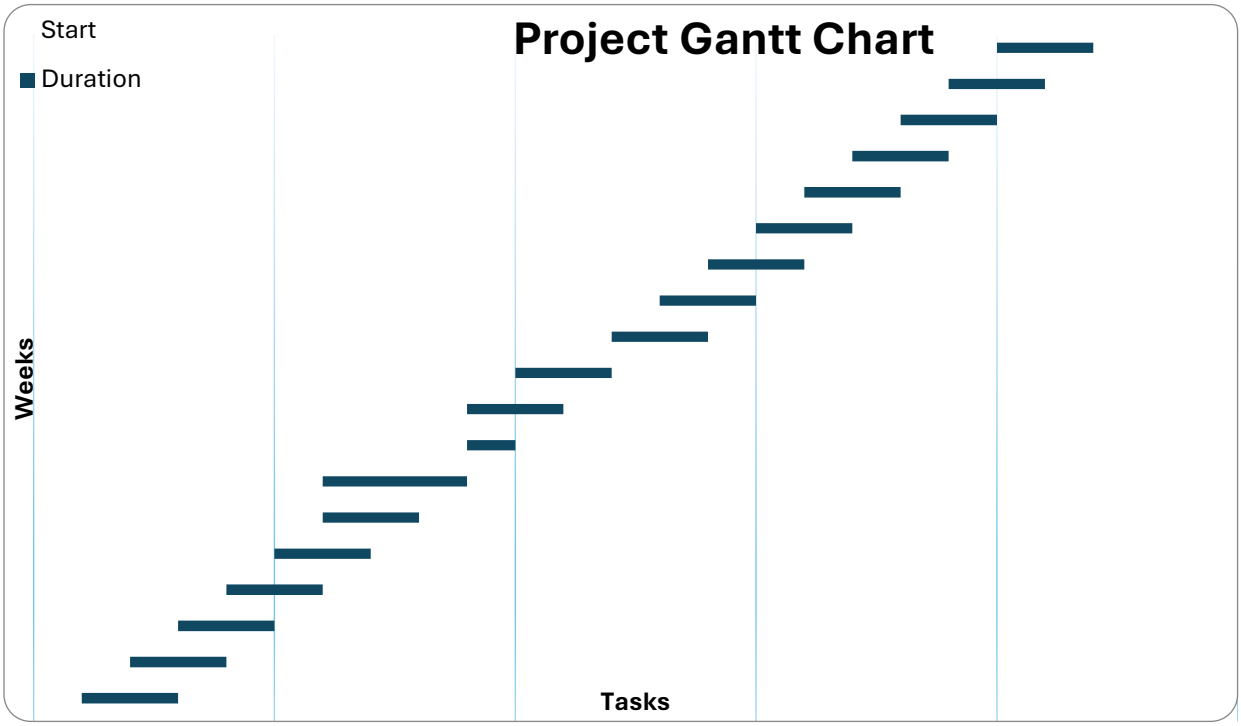


Figure 6: project Gantt Chart.

Table 8: Task Table

Task ID	Task Description	Start Week	Duration (Weeks)	End Week
1.1	In-depth literature review on LCA & textile impacts	1	2	2
1.2	Identify and finalize LCA data sources	2	2	3
1.3	Finalize system requirements and design	3	2	4
2.1	Compile LCA factor database from sources	4	2	5
2.2	Develop a feature engineering pipeline	5	2	6
2.3	Implement baseline (lookup) model	6	2	7
2.4	Implement and train ML regression models (RF, XGB)	6	3	8
3.1	Evaluate models using test set (R^2, MAE, RMSE)	9	1	9
3.2	Select the best model and hyperparameter tuning	9	2	10
3.3	Develop Green Score calculation logic	10	2	11
4.1	Develop a Fast API microservice with endpoints	12	2	13
4.2	Implement input validation and error handling	13	2	14
4.3	Write unit tests and integration tests	14	2	15
4.4	Containerize the application with Docker	15	2	16
5.1	Conduct performance and load testing	16	2	17
5.2	Integrate with the main platform for end-to-end testing	17	2	18
5.3	Finalize API documentation and user guide	18	2	19
6.1	Prepare the final project report section	19	2	20

6.2	Prepare final presentation/demo	20	2	21

10. References

[1] K. Niinimäki, "The environmental price of fast fashion," *Nature Reviews Earth & Environment*, 2020.

[2] E. MacArthur, "A New Textiles Economy: Redesigning Fashion's Future," 2017.

[3] United Nations Environment Programme, "Textile Value Chain," 2020.

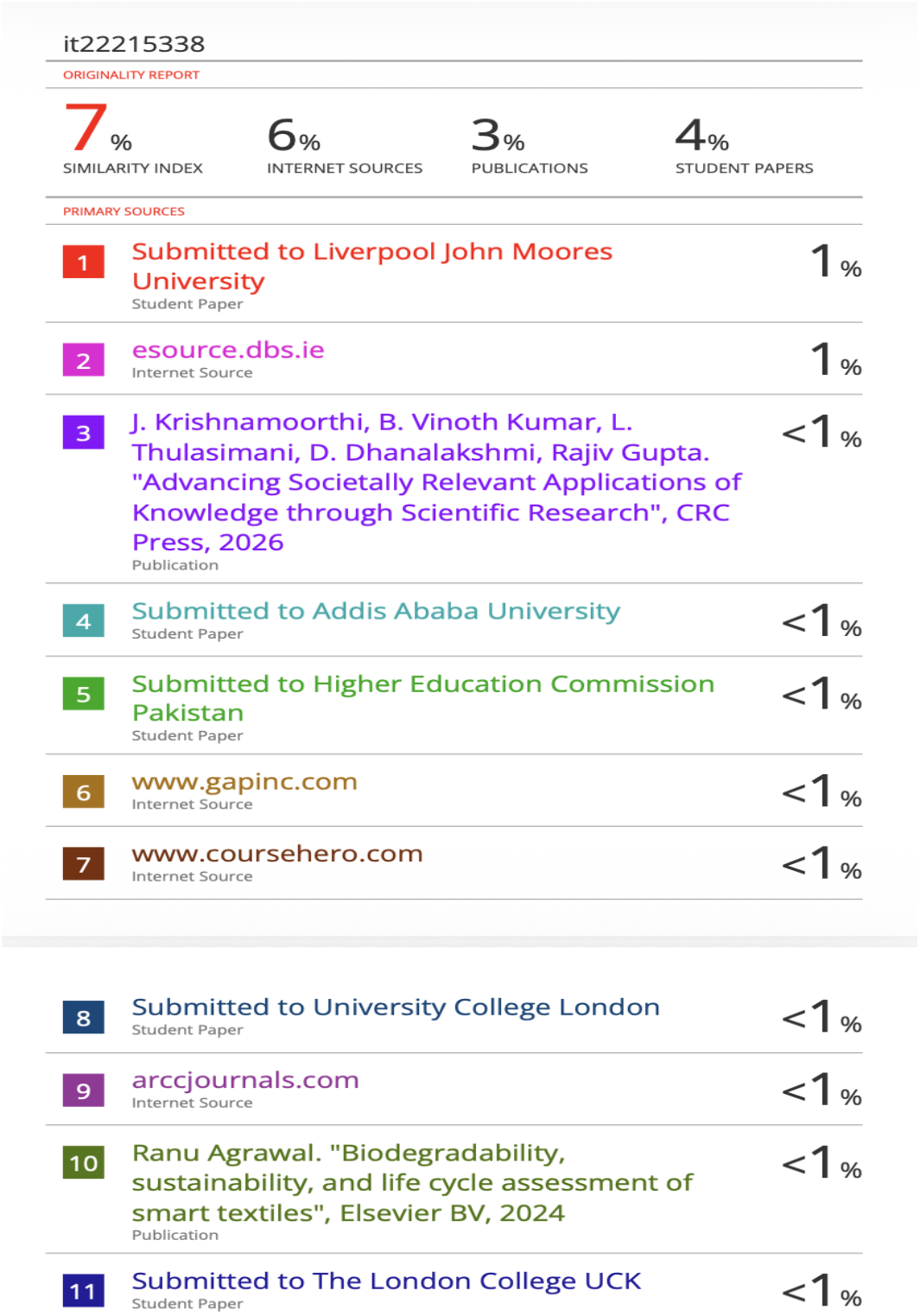
[4] International Organization for Standardization, "Environmental management standards," 2006.

[5] L. Shen, E. Worrell, and M. K. Patel, "Environmental impact assessment of man-made cellulose fibres," 2010.

[6] A. K. Chapagain *et al.*, "The water footprint of cotton consumption," *Ecological Economics*, 2006.

[7] L. Shen, E. Worrell, and M. K. Patel, "Environmental impact assessment of man-made cellulose fibres," *Resources, Conservation and Recycling*, 2010.

11. Appendices



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